DWG. No. 310SC90-776

DWG. No. SC90-776 R0// REGD. 2015-05-29
CHKD. M. SAKAKIBARA/ APPD. M. SAKAKIBARA

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REVISIONS

REV.	DESCRIPTION	REVD.	CHKD.	APPD.	DATE
B	TO REFLECT THE DESIGN PROGRESS AND CUSTOMER COMMENTS	R. KOJIMA	M. SAKAKIBARA	N. MATSUMOTO	2013-12-06
C	TO REFLECT THE DESIGN PROGRESS AND CUSTOMER COMMENTS	R. KOJIMA	M. SAKAKIBARA	N. MATSUMOTO	2014-02-13
D	TO REFLECT THE DESIGN PROGRESS	R. KOJIMA	M. SAKAKIBARA	N. MATSUMOTO	2014-10-02
E	TO REFLECT THE DESIGN PROGRESS	M. SAKAKIBARA	N. MATSUMOTO	N. MATSUMOTO	2014-11-14
0	SUBMISSION AS "FINAL" ENGINEERING DOCUMENT	R. KOJIMA	M. SAKAKIBARA	M. SAKAKIBARA	2015-05-29

ATS CRITERIA LIST AND CRITERIA TABLE

REFERENCE DRAWINGS

1. TURBINE PROTECTION LOGIC DIAGRAM
   MHPS DWG. NO.; 310QC62-580
2. TURBINE GENERATOR INTERLOCK BLOCK DIAGRAM
   MHPS DWG. NO.; 310SC90-774
3. ATS UNIT SEQUENCE CONTROL PROGRAM
   MHPS DWG. NO.; 310SC90-775
4. GENERATOR PROTECTION/BREAKER LOGIC DIAGRAM
   MHPS DWG. NO.; 310SD03-743

三菱日立パワーシステムズ株式会社
2015-05-29
MITSUBISHI HITACHI POWER SYSTEMS, LTD.

FOR FINAL

CONFIDENTIAL

PROJECT TITLE / LOCATION

SMC DAVAO POWER PLANT PROJECT
MALITA, PROVINCE OF DAVAO DEL SUR, PHILIPPINES

OWNER / ADDRESS

San Miguel Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R.O.C.

Drawn : Name/Date	R. KOJIMA	/2013-06-06
Checked : Name/Date	M. SAKAKIBARA	/2013-06-06
Approved: Name/Date	S. NEMOTO	/2013-06-06

Title,supplementary title

ATS CRITERIA LIST AND CRITERIA TABLE

Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
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MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

CAD SOFTWARE: AutoCAD | SIZE DESIGNATION: A3

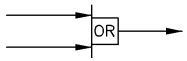
GENERAL

1. ATS has the following seven (8) start-up steps, as shown on "ATS CRITERIA TABLE FOR HP START MODE" AND "ATS CRITERIA TABLE FOR IP START MODE".
   0) HP TURBINE WARMING
   1) TURBINE PREPARATION
   2) TURBINE RUB CHECK
   3) ACCELERATION
   4) EXCITATION
   5) SYNCHRONIZING AND INITIAL LOAD
   6) LOAD UP
   7) NORMAL OPERATION
2. ATS has the following three (3) check function.
   1) STEP IN CONDITION (Ai)
      Check items of the specified turbine and generator conditions before ATS executes the control step criteria (Bi) of number "i", when foregoing control step of number "i-1" is finished completely.
   2) STEP CRITERIA (Bi)
      Check items of the specified turbine and generator conditions continuously, when ATS is executing the control step of number "i".
   3) STEP OUT CONDITION (Di)
      Check items of the specified turbine and generator conditions to decide the starting control step, when ATS is in service on any one control step.
3. Items checked at each step are shown on criteria No. 1.1~2.94. When the main turbine or its auxiliaries do not satisfy the required condition, the lamp of the corresponding unsatisfied item on the operator's console panel is lighted.
4. All analog and digital input signals to ATS are shown on CRITERIA LIST.

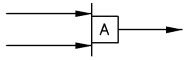
NOTE:

1. "n" INDICATES UNIT NO. AND IT CHANGES ACCORDING TO EACH UNIT.

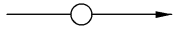
SYMBOL



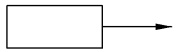
'OR' GATE



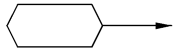
'AND' GATE



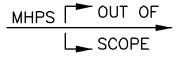
'NOT' GATE



ANALOG INPUT



DIGITAL INPUT



SCOPE OF SUPPLY

ABBREVIATION

TI	···	TURBINE ISLAND
BI	···	BOILER ISLAND
EI	···	ELECTRICAL ISLAND
MHPS	···	MITSUBISHI HITACHI POWER SYSTEMS LTD
TCS	···	TURBINE CONTROL SYSTEM
ATS	···	AUTOMATIC TURBINE STARTUP/SHUTDOWN SYSTEM
ASD	···	AUTOMATIC SYNCHRONIZING DEVICE
AVR	···	AUTOMATIC VOLTAGE REGULATOR
GCB	···	GENERATOR CIRCUIT BREAKER
DISCON	···	DISCONNECTING SWITCH
DCS	···	DISTRIBUTED CONTROL SYSTEM
LLM	···	LOAD LIMIT

TYPE	–				PAGE	2
SERVICE	GENERAL, SYMBOL & ABBREVIATION					
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MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

ATS CRITERIA TABLE FOR IP START MODE

NO.	STEP / CRITERIA NAME	0			1			2			3			4			5			6			7			8			HOLD
		HP TBN WARMING			PREPARA-TION			TURBINE RUB CHECK			ACCELERA-TION			EXCITATION			SYNCH. & INIT. LOAD			LOAD UP			NORMAL OPERATION			TURBINE SHUTDOWN			
		A0	B0	D0	A1	B1	D1	A2	B2	D2	A3	B3	D3	A4	B4	D4	A5	B5	D5	A6	B6	D6	A7	B7	D7	A8	B8	D8	
1.1	TSI OK					○		○	○		○	○		○	○		○	○		○	○			○					
1.2	TCS NORMAL					○		○	○		○	○		○	○		○	○		○	○			○					
1.3	MAIN STEAM PRESS. NOT LOW					○		○	○		○	○		○	○		○	○		○	○			○					
1.4	MAIN STEAM TEMP. NOT LOW					○		○	○		○	○		○	○		○	○		○	○			○					
1.5	REHEAT STEAM PRESS. NOT LOW					○		○	○		○	○		○	○		○	○											
1.6	REHEAT STEAM TEMP. NOT LOW					○		○	○		○	○		○	○		○	○											
1.71	MAIN STEAM TEMP. MISMATCH OK							○			○																		
1.72	REHEAT STEAM TEMP. MISMATCH OK							○			○																		
1.8	HP/LP BYPASS CONTROL IN AUTO MODE					○		○	○		○	○		○	○		○	○		○	○								
1.9																													
1.10	GLAND STEAM SEAL HEADER PRESS. NOT LOW	○	○			○		○	○		○	○		○	○		○	○		○	○			○					
1.11	CONDENSER VACUUM OK	○	○			○		○	○		○	○		○	○		○	○		○	○			○					○
1.12	LP TURBINE EXHAUST HOOD STEAM TEMP. NOT HIGH					○		○	○		○	○		○	○		○	○		○	○			○					○
1.13																													
1.14	CONTROL OIL PRESS. NOT LOW					○		○	○		○	○		○	○		○	○		○	○			○					
1.15	BEARING OIL PRESS. NOT LOW	○	○			○		○	○		○	○		○	○		○	○		○	○			○					○
1.16	BEARING OIL TEMP. OK	○	○			○		○	○		○	○		○	○		○	○		○	○			○					
1.17	MAIN OIL TANK LEVEL OK					○		○	○		○	○		○	○		○	○		○	○			○					
1.18	MAIN OIL TANK VAPOUR EXTRACTOR RUNNING	○	○			○		○	○		○	○		○	○		○	○		○	○			○					
1.19	TURBINE TURNING	○	○			○		○																					
1.20	TURBINE RESET					△	○	○	○		○	○		○	○		○	○		○	○			○					
1.21	MSV OPEN					△	○	○	○		○	○		○	○		○	○		○	○			○					
1.22	RSV OPEN						○	○	○		○	○		○	○		○	○		○	○			○					
1.23	LOAD LIMITER UPPER					△	○	○	○		○	○		○	○		○	○		○									
1.24	VIBRATION NOT HIGH					○		○	○		○	○		○	○		○	○		○	○			○					○
1.25	DRAIN VALVE OPEN						○	○	○		○	○		○	○		○	○		○									
1.26																													
1.27	ECCENTRICITY NOT HIGH					○		○	○																				
1.28	DIFF. EXPANSION OK					○		○	○		○	○		○	○		○	○		○	○			○					○
1.29	ROTOR POSITION OK					○		○	○		○	○		○	○		○	○		○	○			○					
1.30	BEARING METAL TEMP. NOT HIGH					○		○	○		○	○		○	○		○	○		○	○			○					
1.31	THRUST METAL TEMP. NOT HIGH					○		○	○		○	○		○	○		○	○		○	○			○					
1.32	MSV CHEST TEMP. DIFF. OK					○		○	○		○	○		○	○		○	○		○	○			○					
1.33	CV CHEST TEMP. DIFF. OK					○	○	○	○		○	○		○	○		○	○		○	○			○					
1.34	HP FIRST STAGE TEMP. DIFF. OK					○		○	○		○	○		○	○		○	○		○	○			○					
1.35	REHEAT BOWL TEMP. DIFF. OK					○		○	○		○	○		○	○		○	○		○	○			○					
1.36																													
1.37																													
1.38																													
1.39																													
1.40	GENERATOR TEMPERATURE	○	○			○		○	○		○	○		○	○		○	○		○	○			○					
1.41	STEAM TURBINE TRIP DEMAND DUE TO ELECTRICAL TROUBLE NOT					○		○	○		○	○		○	○		○	○		○	○			○					
1.42	AVR OK															○	○	○		○	○			○					
1.43	AVR IN AUTO															○	○	○		○	○			○					
1.44	41 CLOSED															○	○	○		○	○			○					
1.45	GEN. VOLTAGE OK															○	○	○		○	○			○					
1.46	ASD IN USE																○	○											
1.47	GEN. DISCON CLOSED													○	○	○	○	○		○	○			○					
1.48	52G CLOSED																	△	○	○	○			○					
1.49	52G OPENED																										△	○	
1.50	41 OPENED																										△	○	
1.51	TURBINE TRIPPED																										△	○	
1.52	52L CLOSED																	△											
1.53																													
2.1	SHELL EXPANSION OK											△																	
2.2																													
2.3	TURBINE RUBCHECK COMPLETE SPEED									○	○																		
2.4	TURBINE RATED SPEED												○	○	○		○												
2.5	GEN. LOAD > INITIAL LOAD																		○	○									
2.6	IP TO HP TRANSFER COMPLETE																					○	○						
2.7	V/F (START UP)													○	○														
2.11	TURBINE SHUTDOWN LOAD																										△		
2.81	CV CHEST WARMING MODE SET					△																							
2.82	CV CHEST WARMING AUTO MODE					△																							
2.83	CV CHEST WARMING START					△																							
2.84	CV CHEST WARMING STOP					△																							
2.85	CV CHEST WARMING MODE RESET					△																							
2.91																													
2.92	HP TURBINE WARMING COMPLETE			○	○																								
2.94																													

△ : USED ONLY FOR SEQUENTIAL ACTION

○ : CRITERIA MONITORING TIMING

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TYPE –

SERVICE

CRITERIA TABLE

Date of issue	Rev.	Sheet
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Scale 1 : 1

Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

ATS CRITERIA TABLE FOR IP START MODE

NO.	STEP / CRITERIA NAME	0			1			2			3			4			5			6			7			8			HOLD
		HP TBN WARMING			PREPARA-TION			TURBINE RUB CHECK			ACCELERA-TION			EXCITATION			SYNCH. & INIT. LOAD			LOAD UP			NORMAL OPERATION			TURBINE SHUTDOWN			
		A0	B0	D0	A1	B1	D1	A2	B2	D2	A3	B3	D3	A4	B4	D4	A5	B5	D5	A6	B6	D6	A7	B7	D7	A8	B8	D8	
3.1	HOUSE LOAD OPERATION JUDGEMENT																	○											

△ : USED ONLY FOR SEQUENTIAL ACTION

○ : CRITERIA MONITORING TIMING

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TYPE –

SERVICE PURPOSE, ABBREVIATIONS, AND SYMBOLS

Responsible dep.: Hitachi I&C Engineering Section/ Instrument & Control Department

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Rev. 0

Sheet 4

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MITSUBISHI HITACHI POWER SYSTEMS, LTD.

Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-776

CRITERIA NO.	CRITERIA NAME	CONTACT COMBINATION	CONTACT COMBINATION & SET POINT	CONTACT SUPPLY	IBD SH-NO.	REMARKS
1.1	TSI OK	TURBINE SUPERVISORY INSTRUMENT TROUBLE → ATS	ON: TURBINE SUPERVISORY INSTRUMENT POWER IN SERVICE AND INSTRUMENT NOT IN TROUBLE	MHPS	TD-4	
1.2	TCS NORMAL	TCS TCS MINOR TROUBLE → ATS	ON: TCS NORMAL	MHPS	TD-1	
1.3	MAIN STEAM PRESS. NOT LOW	PT-MSn901-1,2 MSV INLET STEAM PRESS. → ATS	ON: MAIN STEAM PRESS. NOT LESS THAN	MHPS	———	"MSV INLET STEAM PRESS." SIGNAL INPUT IS SHOWN ON SH.16 ITEM NO.1 OF THIS DRAWING
1.4	MAIN STEAM TEMP. NOT LOW	TE-MSn902-1,2,3 MSV INLET STEAM TEMP. → ATS	ON: MAIN STEAM TEMP. NOT LESS THAN	MHPS	———	"MSV INLET STEAM TEMP." SIGNAL INPUT IS SHOWN ON SH.16 ITEM NO.2 OF THIS DRAWING
1.5	REHEAT STEAM PRESS. NOT LOW	PT-HRn901 HOT REHEAT STEAM PRESS. → ATS	ON: REHEAT STEAM PRESS. NOT LESS THAN	MHPS	———	"HOT REHEAT STEAM PRESS" SIGNAL INPUT IS SHOWN ON SH.16 ITEM NO.3 OF THIS DRAWING
1.6	REHEAT STEAM TEMP. NOT LOW	TE-HRn902 HOT REHEAT STEAM TEMP. → ATS	ON: REHEAT STEAM TEMP. NOT LESS THAN	MHPS	———	"HOT REHEAT STEAM TEMP" SIGNAL INPUT IS SHOWN ON SH.16 ITEM NO.4 OF THIS DRAWING
1.71	MAIN STEAM TEMP. MISMATCH OK	TE-MSn902-4 MSV INLET STEAM TEMP. TE-STn001-1 HP TURBINE FIRST STAGE INNER WALL METAL TEMP. → ATS	ON: DIFF. BETWEEN MSV INLET STEAM TEMP. AND HP TURBINE FIRST STAGE INNER WALL METAL TEMP. IS CALCULATED IN ATS. SETPOINT: REFER TO SH.21 FIG.2 ALLOWABLE SETPOINT BETWEEN UPPER LIMIT AND TARGET TEMP.	MHPS	———	"MSV INLET STEAM TEMP" SIGNAL INPUT IS SHOWN ON SH.16 ITEM NO.2 OF THIS DRAWING AND ALSO "HP TURBINE FIRST STAGE INNER WALL METAL TEMP" SIGNAL INPUT IS SHOWN ON SH.16 ITEM NO.5
1.72	REHEAT STEAM TEMP. MISMATCH OK	TE-HRn902 HOT REHEAT STEAM TEMP. TE-STn004-1,2 REHEAT BOWL INNER WALL METAL TEMP. → ATS	ON: DIFF. BETWEEN REHEAT STEAM TEMP. AND REHEAT BOWL INNER WALL METAL TEMP. IS CALCULATED IN ATS. SETPOINT: REFER TO SH.21 FIG.3 ALLOWABLE SETPOINT BETWEEN UPPER LIMIT AND TARGET TEMP.	MHPS	———	"HOT REHEAT STEAM TEMP" SIGNAL INPUT IS SHOWN ON SH.16 ITEM NO.4 OF THIS DRAWING AND ALSO "REHEAT BOWL INNER WALL METAL TEMP" SIGNAL INPUT IS SHOMN ON SH.16 ITEM NO.7

TYPE	–
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SERVICE	CRITERIA LIST

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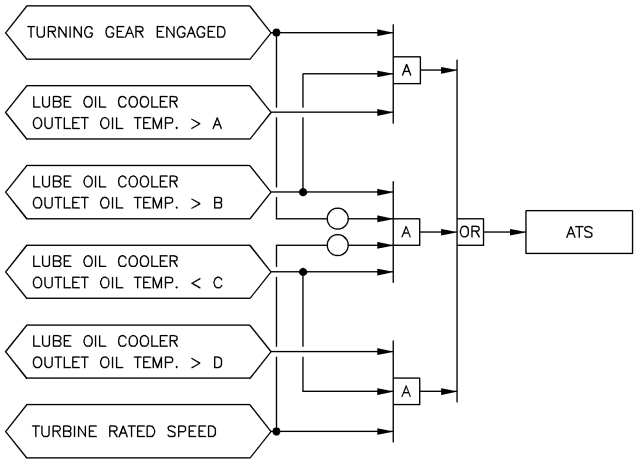
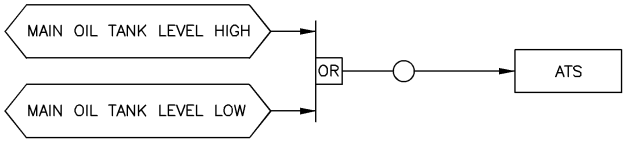
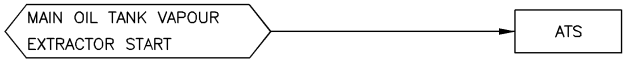
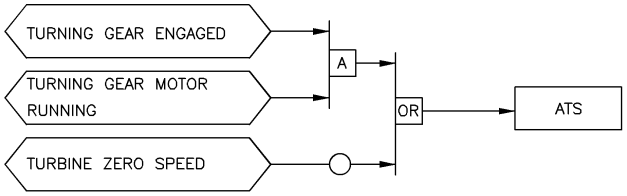
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Engineering Headquarters / Plant Engineering Division

CRITERIA NO.	CRITERIA NAME	CONTACT COMBINATION	CONTACT COMBINATION & SET POINT	CONTACT SUPPLY	IBD SH–NO.	REMARKS
1.8	HP/LP BYPASS CONTROL IN AUTO MODE	HP BYPASS CONTROL IN AUTO LP BYPASS CONTROL IN AUTO → A → FHI → ATS	ON: HP/LP BYPASS CONTROL IN AUTO MODE	FHI	——	
1.10	GLAND STEAM SEAL HEADER PRESS. NOT LOW	GLAND STEAM SEAL HEADER PRESS. (A) GLAND STEAM SEAL HEADER PRESS. (B) → SEL → CMP → ATS	ON: GLAND STEAM SEAL PRESS. NORMAL SETPOINT : REFER TO IBD	MHPS	TD–3	
1.11	CONDENSER VACUUM OK	CONDENSER VACUUM LOW → ATS	ON: CONDENSER VACUUM PRESS. NORMAL SETPOINT : REFER TO IBD	MHPS	TA–10	
1.12	LP TURBINE EXHAUST HOOD STEAM TEMP. NOT HIGH	LP TURBINE EXHAUST HOOD STEAM TEMP. HIGH → ATS	ON: LP TURBINE EXHAUST HOOD STEAM TEMP. NORMAL SETPOINT : REFER TO IBD	MHPS	TA–14	
1.13						
1.14	CONTROL OIL PRESS. NOT LOW	CONTROL OIL PRESS. LOW → ATS	ON: CONTROL OIL PRESS. NORMAL SETPOINT : REFER TO IBD	MHPS	TA–13	
1.15	BEARING OIL PRESS. NOT LOW	BEARING OIL PRESS. LOW → ATS	ON: BEARING OIL PRESS. NORMAL SETPOINT : REFER TO IBD	MHPS	TA–13	

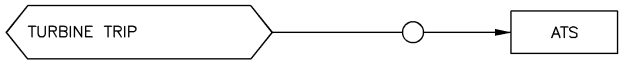
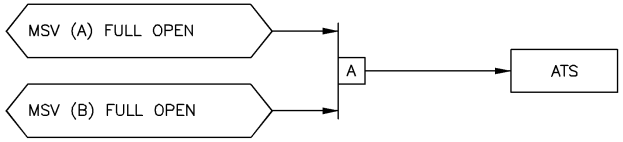
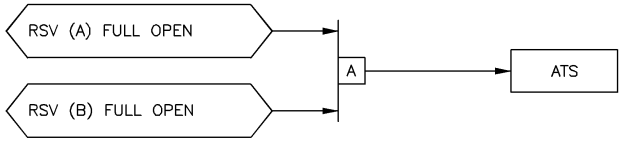
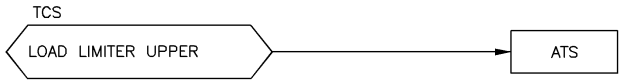
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CRITERIA NO.	CRITERIA NAME	CONTACT COMBINATION	CONTACT COMBINATION & SET POINT	CONTACT SUPPLY	IBD SH-NO.	REMARKS
1.16	BEARING OIL TEMP. OK		ON: TURBINE OIL COOLER OUTLET OIL TEMP. NORMAL SETPOINT: REFER TO IBD	MHPS	TD-1	
1.17	MAIN OIL TANK LEVEL OK		ON: MAIN OIL TANK LEVEL NORMAL SETPOINT: REFER TO IBD	MHPS	TD-3	
1.18	MAIN OIL TANK VAPOUR EXTRACTOR RUNNING		ON: MAIN OIL TANK VAPOUR EXTRACTOR START	MHPS	TB-12	
1.19	TURBINE TURNING		ON: TURNING GEAR MOTOR RUNNING AND TURNING GEAR ENGAGE OR TURBINE NOT ZERO SPEED	MHPS	TB-9	

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CRITERIA NO.	CRITERIA NAME	CONTACT COMBINATION	CONTACT COMBINATION & SET POINT	CONTACT SUPPLY	IBD SH-NO.	REMARKS
1.20	TURBINE RESET		ON: TURBINE IS NOT TRIP	MHPS	TA-1	
1.21	MSV OPEN		ON: BOTH MSV FULL OPEN	MHPS	TA-32	
1.22	RSV OPEN		ON: BOTH RSV FULL OPEN	MHPS	TA-37	
1.23	LOAD LIMITER UPPER		ON: LOAD LIMITER UPPER	MHPS	———	

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1.24	VIBRATION NOT HIGH	NO.1 TURBINE SHAFT VIBRATION (X) HIGH NO.1 TURBINE SHAFT VIBRATION (Y) HIGH NO.2 TURBINE SHAFT VIBRATION (X) HIGH NO.2 TURBINE SHAFT VIBRATION (Y) HIGH NO.3 GEN. SHAFT VIBRATION (X) HIGH NO.3 GEN. SHAFT VIBRATION (Y) HIGH NO.4 GEN. SHAFT VIBRATION (X) HIGH NO.4 GEN. SHAFT VIBRATION (Y) HIGH OR → ATS	ON: TURBINE & GEN. SHAFT VIBRATION NORMAL SETPOINT: REFER TO IBD	MHPS	TA-18	
1.25	DRAIN VALVE OPEN	MSV(A) ABOVE SEAT SECONDARY DRAIN VALVE FULL OPEN MSV(B) ABOVE SEAT SECONDARY DRAIN VALVE FULL OPEN CV(A) ABOVE SEAT SECONDARY DRAIN VALVE FULL OPEN CV(B) ABOVE SEAT SECONDARY DRAIN VALVE FULL OPEN CRV(A) BELOW SEAT SECONDARY DRAIN VALVE FULL OPEN CRV(B) BELOW SEAT SECONDARY DRAIN VALVE FULL OPEN A → ATS	ON: ALL DRAIN VALVE OPEN	MHPS	TC-1	
1.26						

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1.27	ECCENTRICITY NOT HIGH	LOW SPEED ECCENTRICITY DETECTOR HIGH → ATS	ON: LOW SPEED ECCENTRICITY DETECTOR NOT HIGH SETPOINT: REFER TO IBD	MHPS	TD-4	
1.28	DIFF. EXPANSION OK	LP TURBINE DIFF. EXPANSION (A) (ROTOR LONG) LP TURBINE DIFF. EXPANSION (B) (ROTOR SHORT) → A → ATS	ON: DIFF. EXPANSION OK SETPOINT: REFER TO IBD	MHPS	TD-4	
1.29	ROTOR POSITION OK	THRUST BEARING FAILURE (FRONT) THRUST BEARING FAILURE (REAR) → A → ATS	ON: ROTOR POSITION NORMAL SETPOINT: REFER TO IBD	MHPS	TA-12	
1.30	BEARING METAL TEMP. NOT HIGH	NO.1 TURBINE JOURNAL BEARING METAL TEMP HIGH NO.2 TURBINE JOURNAL BEARING METAL TEMP HIGH NO.3 BEARING METAL TEMP HIGH NO.4 BEARING METAL TEMP HIGH → A → ATS	ON: TURBINE & GEN. BEARING METAL TEMP. NORMAL SETPOINT: REFER TO IBD	MHPS	TD-2	

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1.31	THRUST BEARING METAL TEMP. NOT HIGH	THRUST BEARING FRONT-END EMBEDDED METAL TEMP. 1 HIGH THRUST BEARING FRONT-END EMBEDDED METAL TEMP. 2 HIGH THRUST BEARING REAR-END EMBEDDED METAL TEMP. 1 HIGH THRUST BEARING REAR-END EMBEDDED METAL TEMP. 2 HIGH → A → ATS	ON: ALL THRUST BEARING METAL TEMP. NOT HIGH SETPOINT: REFER TO IBD	MHPS	TD-2	
1.32	MSV CHEST TEMP. DIFF OK	TE-STn903-B1 MSV (B) INNER WALL METAL TEMP. TE-STn903-B2 MSV (B) OUTER WALL METAL TEMP. → ATS	ON: DIFF. BETWEEN MSV CHEST INNER WALL METAL TEMP. AND OUTER WALL METAL TEMP. IS CALCULATED IN ATS. SETPOINT: REFER TO SH.22 FIG.4	MHPS	———	"MSV (B) INNER WALL METAL TEMP" SIGNAL INPUT IS SHOWN ON SH.17 ITEM NO.13 OF THIS DRAWING AND ALSO "MSV (B) OUTER WALL METAL TEMP" SIGNAL INPUT IS SHOWN ON SH.17 ITEM NO.14
1.33	CV CHEST TEMP. DIFF. OK	TE-STn904-B1 CV (B) INNER WALL METAL TEMP. TE-STn904-B2 CV (B) OUTER WALL METAL TEMP. → ATS	ON: DIFF. BETWEEN CV CHEST INNER WALL METAL TEMP. AND OUTER WALL METAL TEMP. IS CALCULATED IN ATS. SETPOINT: REFER TO SH.22 FIG.5	MHPS	———	"CV (B) INNER WALL METAL TEMP" SIGNAL INPUT IS SHOWN ON SH.17 ITEM NO.18 OF THIS DRAWING AND ALSO "CV (B) OUTER WALL METAL TEMP" SIGNAL INPUT IS SHOWN ON SH.17 ITEM NO.20
1.34	HP TURBINE FIRST STAGE TEMP. DIFF. OK	TE-STn001-1 HP TURBINE FIRST STAGE INNER WALL METAL TEMP. TE-STn001-2 HP TURBINE FIRST STAGE OUTER WALL METAL TEMP. → ATS	ON: DIFF. BETWEEN HP TURBINE FIRST STAGE INNER WALL METAL TEMP. AND OUTER WALL METAL TEMP. IS CALCULATED IN ATS. SETPOINT: REFER TO SH.23 FIG.6	MHPS	———	"HP TURBINE FIRST STAGE INNER WALL METAL TEMP" SIGNAL INPUT IS SHOWN ON SH.16 ITEM NO.5 OF THIS DRAWING AND ALSO "HP TURBINE FIRST STAGE OUTER WALL METAL TEMP" SIGNAL INPUT IS SHOWN ON SH.16 ITEM NO.6

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1.35	REHEAT BOWL TEMP. DIFF. OK	TE-STn004-1,2 REHEAT BOWL INNER WALL METAL TEMP. TE-STn005 REHEAT BOWL OUTER WALL METAL TEMP. → ATS	ON: DIFF. BETWEEN REHEAT BOWL INNER WALL METAL TEMP. AND OUTER WALL METAL TEMP. IS CALCULATED IN ATS. SETPOINT: REFER TO SH.23 FIG.7	MHPS	———	"REHEAT BOWL INNER WALL METAL TEMP" SIGNAL INPUT IS SHOWN ON SH.16 ITEM NO.7 OF THIS DRAWING AND ALSO "REHEAT BOWL OUTER WALL METAL TEMP" SIGNAL INPUT IS SHOWN ON SH.16 ITEM NO.8
1.36						
1.37						
1.38						
1.39						
1.40	GENERATOR TEMPERATURE	GENERATOR TEMPERATURE TROUBLE → ATS	ON: GENERATOR TEMPERATURE NOT IN TROUBLE SETPOINT: REFER TO IBD	MHPS	GD-5	
1.41	STEAM TURBINE TRIP DEMAND DUE TO ELECTRICAL TROUBLE NOT	STEAM TURBINE TRIP DEMAND DUE TO ELECTRICAL TROUBLE → ATS	ON: STEAM TURBINE TRIP DEMAND DUE TO ELECTRICAL TROUBLE NOT	MHPS	TA-9	
1.42	AVR OK	AVR MINOR TROUBLE → ATS	ON: AVR NOT IN MINOR TROUBLE	MHPS	G-4	

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1.43	AVR IN AUTO	AVR IN AUTO → ATS	ON: AVR IN AUTO	MHPS	G-4	
1.44	41 CLOSED	41 CLOSED → ATS	ON: 41 CLOSED	MHPS	G-4	
1.45	GEN. VOLTAGE OK	GEN. VOLTAGE OK → ATS	ON: GEN. VOLTAGE OK	MHPS	G-1B	
1.46	ASD IN USE	ASD IN USE → ATS	ON: ASD IN USE	MHPS	G-1B	
1.47	GEN DISCON CLOSED	GEN. DISCON CLOSED → ATS	ON: GEN. DISCON CLOSED	MHPS	G-1A	
1.48	52G CLOSED	52G CLOSED → ATS	ON: 52G CLOSED	MHPS	G-1A	
1.49	52G OPENED	52G OPENED → ATS	ON: 52G OPENED	MHPS	G-1A	
1.50	41 OPENED	41 OPENED → ATS	ON: 41 OPENED	MHPS	G-4	
1.51	TURBINE TRIPPED	TURBINE RESET ○→ ATS	ON: TURBINE RESET	MHPS	TA-1	
1.52	52L CLOSED	52L CLOSED → ATS	ON: 52L CLOSED	MHPS	G-2A	
1.53						

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2.1	SHELL EXPANSION OK	TSI ZE-STn001 TURBINE SHELL EXPANSION (HP TURBINE SIDE) NORMAL → ATS	ON: TURBINE SHELL EXPANSION	MHPS	TD-4	
2.3	TURBINE RUB CHECK COMPLETE SPEED	THE LOGIC IS SHOWN ON FIG.1 IN THIS SHEET.	ON: TURBINE RUB CHECK COMPLETE SPEED SETPOINT: REFER TO FIG.1 ON THIS SHEET.	MHPS	———	
2.4	TURBINE RATED SPEED	TCS SE-STn001-1,2,3 TURBINE SPEED → ATS	ON: TURBINE SPEED MORE THAN 3550rpm	MHPS	———	
2.5	GEN. LOAD > INITIAL LOAD	TCS GEN. LOAD → ATS	ON: GEN. LOAD MORE THAN INITIAL LOAD	MHPS	———	
2.6	IP TO HP TRANSFER COMPLETE	TCS IP TO HP TRANSFER COMPLETE → ATS	ON: TRANSFER REGION COMPLETE	MHPS	———	
2.7	V/F (START UP)	V/F (START UP) → S(R) → ATS TCS ATS OFF → OR → S(R) TCS TURBINE TRIP → OR	ON: V/F (START UP) NOT OPERATED	MHPS	G-8A	
2.11	TURBINE SHUTDOWN LOAD	TCS GEN. LOAD → ATS	ON: GEN. LOAD LESS THAN 20%	MHPS	———	

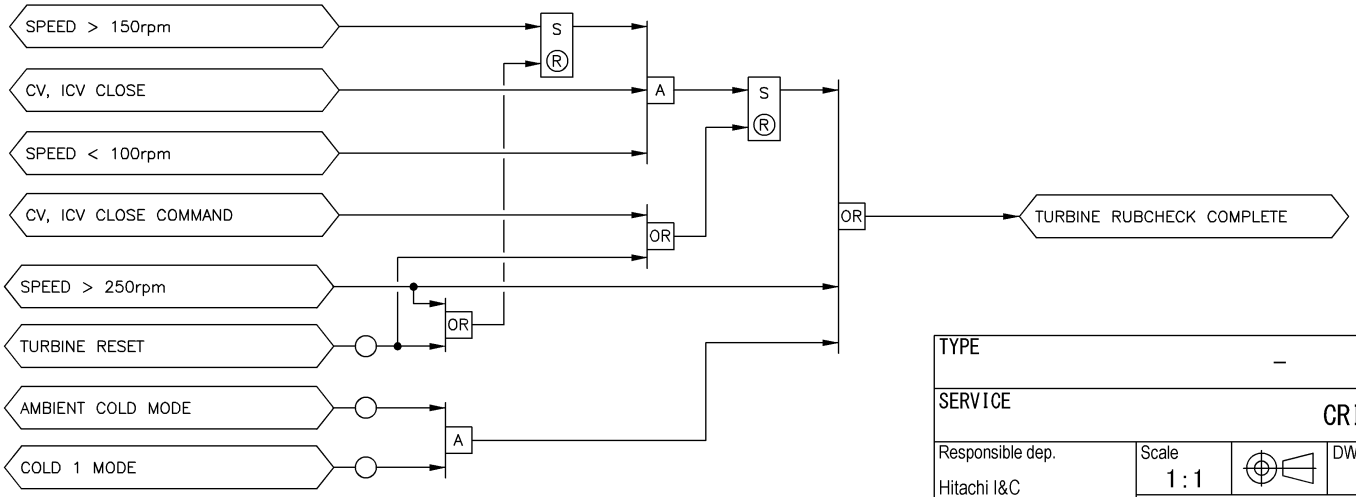


FIG.1 TURBINE RUBCHECK COMPLETE LOGIC

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2.81	CV CHEST WARMING MODE SET	TCS: CV CHEST WARMING MODE SET → ATS	ON: CV CHEST WARMING MODE SET	MHPS	———	
2.82	CV CHEST WARMING AUTO MODE	TCS: CV CHEST WARMING AUTO MODE → ATS	ON: CV CHEST WARMING AUTO MODE	MHPS	———	
2.83	CV CHEST WARMING VALVE START	TCS: CV CHEST WARMING START → ATS	ON: CV CHEST WARMING START	MHPS	———	
2.84	CV CHEST WARMING STOP	TCS: CV CHEST WARMING STOP → ATS	ON: CV CHEST WARMING STOP	MHPS	———	
2.85	CV CHEST WARMING MODE RESET	CV CHEST WARMING MODE RESET → ATS	ON: CV CHEST WARMING MODE RESET	MHPS	———	
2.91						
2.92	HP TURBINE WARMING COMPLETE	HP TURBINE WARMING COMPLETE → ATS	ON: HP TURBINE WARMING COMPLETE	MHPS	TE-1	
3.1	HOUSE LOAD OPERATION JUDGEMENT	52G CLOSE, 52L OPEN → A → ATS	ON: 52G CLOSED AND 52L OPENED (HOUSE LOAD OPERATION)	MHPS	———	

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ANALOG INPUT

NO.	INPUT ITEMS	ANALOG INPUT	REMARKS
1	MAIN STEAM PRESS.	PT-MSn901-1,2 MSV INLET STEAM PRESS. → ATS	
2	MAIN STEAM TEMP.	TE-MSn902-1,2,3,4 MSV INLET STEAM TEMP. → ATS	
3	REHEAT STEAM PRESS.	PT-HRn901 HOT REHEAT STEAM PRESS. → ATS	
4	REHEAT STEAM TEMP.	TE-HRn902 HOT REHEAT STEAM TEMP. → ATS	
5	HP TURBINE FIRST STAGE INNER WALL METAL TEMP.	TE-STn001-1 HP TURBINE FIRST STAGE INNER WALL METAL TEMP. → ATS	
6	HP TURBINE FIRST STAGE OUTER WALL METAL TEMP.	TE-STn001-2 HP TURBINE FIRST STAGE OUTER WALL METAL TEMP. → ATS	
7	REHEAT BOWL INNER WALL METAL TEMP.	TE-STn004-1,2 REHEAT BOWL INNER WALL METAL TEMP. → ATS	

ATS: THERMAL STRESS CALCULATION

NO.	INPUT ITEMS	ANALOG INPUT	REMARKS
8	REHEAT BOWL OUTER WALL METAL TEMP.	TE-STn005 REHEAT BOWL OUTER WALL METAL TEMP. → ATS	
9	GENERATOR MW	TCS GENERATOR MW → ATS	
10	TURBINE SPEED	TCS SE-STn001-1,2,3 TURBINE SPEED → ATS	

ATS: THERMAL STRESS CALCULATION

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ANALOG INPUT

NO.	INPUT ITEMS	ANALOG INPUT	REMARKS
11			
12			
13	MSV (B) INNER WALL METAL TEMP.	TE-STn903-B1 MSV (B) INNER WALL METAL TEMP. → MONITORING	
14	MSV (B) OUTER WALL METAL TEMP.	TE-STn903-B2 MSV (B) OUTER WALL METAL TEMP. → MONITORING	

NO.	INPUT ITEMS	ANALOG INPUT	REMARKS
15			
16			
17			
18	CV (B) CHEST INNER WALL METAL TEMP.	TE-STn904-B1 CV (B) INNER WALL METAL TEMP. → MONITORING	
19			
20	CV (B) CHEST OUTER WALL METAL TEMP.	TE-STn904-B2 CV (B) OUTER WALL METAL TEMP. → MONITORING	

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ATS OUTPUT

NO.	OUTPUT ITEMS	SIGNAL TYPE	REMARKS	NO.	OUTPUT ITEMS	SIGNAL TYPE	REMARKS
1	SET TARGET SPEED TO RPM (RUB CHECK)	COMMUNICATION LINK	ATS → TCS	10	HEAT SOAK OFF	COMMUNICATION LINK	ATS → TCS
2	SET TARGET SPEED TO 1800RPM	COMMUNICATION LINK	ATS → TCS	11	SET INITIAL LOAD	COMMUNICATION LINK	ATS → TCS
3				12	LOAD UP START	COMMUNICATION LINK	ATS → TCS
4	SET CV, ICV CLOSE	COMMUNICATION LINK	ATS → TCS	13	EXCITATION START	MOMENTARY ON CONTACT	ATS → AVR
5	SET ACCELERATION RATE 120 RPM/MIN	COMMUNICATION LINK	ATS → TCS	14	ASD START (52G)	MOMENTARY ON CONTACT	ATS → ASD
6	SET ACCELERATION RATE RPM/MIN (RUB CHECK)	COMMUNICATION LINK	ATS → TCS	15	ASD START (52L)	MOMENTARY ON CONTACT	ATS → ASD
7	SET ACCELERATION RATE 180 RPM/MIN	COMMUNICATION LINK	ATS → TCS	16			
8	SET ACCELERATION RATE 360 RPM/MIN	COMMUNICATION LINK	ATS → TCS	17			
9	HEAT SOAK ON	COMMUNICATION LINK	ATS → TCS	18			

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ATS OUTPUT

NO.	OUTPUT ITEMS	SIGNAL TYPE	REMARKS	NO.	OUTPUT ITEMS	SIGNAL TYPE	REMARKS
19				28	CV CHEST WARMING MODE SET	COMMUNICATION LINK	ATS → TCS
20	SPEED HOLD	COMMUNICATION LINK	ATS → TCS	29	CV CHEST WARMING AUTO MODE	COMMUNICATION LINK	ATS → TCS
21	SPEED RESET	COMMUNICATION LINK	ATS → TCS	30	CV CHEST WARMING START	COMMUNICATION LINK	ATS → TCS
22	THERMAL STRESS (HP SURFACE)	COMMUNICATION LINK	ATS → TCS	31	CV CHEST WARMING STOP	COMMUNICATION LINK	ATS → AVR
23	THERMAL STRESS (IP SURFACE)	COMMUNICATION LINK	ATS → TCS	32	CV CHEST WARMING MODE RESET	COMMUNICATION LINK	ATS → ASD
24	HP TURBINE WARMING START	COMMUNICATION LINK	ATS → TCS	33	TURBINE TRIP	COMMUNICATION LINK	ATS → ASD
25	SPEED SET 3600RPM	COMMUNICATION LINK	ATS → TCS	34			
26	LOAD LIMIT UPPER	COMMUNICATION LINK	ATS → TCS	35			
27	TURBINE RESET	COMMUNICATION LINK	ATS → TCS	36			

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ATS OUTPUT

NO.	OUTPUT ITEMS	SIGNAL TYPE	REMARKS	NO.	OUTPUT ITEMS	SIGNAL TYPE	REMARKS
37	ST AMBIENT COLD START-UP MODE	DRY CONTACT	ATS → FHI → DCS	46			
38	ST COLD 1 START-UP MODE	DRY CONTACT	ATS → FHI → DCS	47			
39	ST COLD 2 START-UP MODE	DRY CONTACT	ATS → FHI → DCS	48			
40	ST WARM START-UP MODE	DRY CONTACT	ATS → FHI → DCS	49			
41	ST HOT START-UP MODE	DRY CONTACT	ATS → FHI → DCS	50			
42	LOAD HOLD	DRY CONTACT	ATS → FHI → DCS	51			
43	MAX TURBINE LOAD UP RATE SET POINT	ANALOG	ATS → FHI → DCS	52			
44				53			
45				54			

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FIG. 2 ALLOWABLE TEMPERATURE DIFFERENCE ON MAIN STEAM AND TURBINE METAL

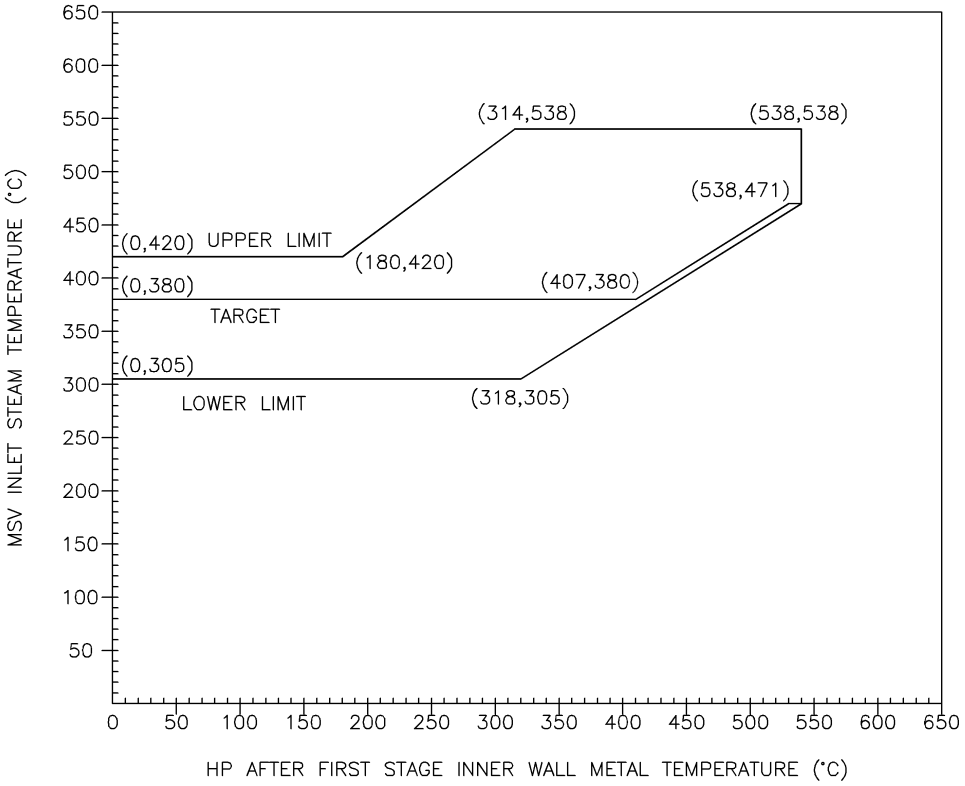
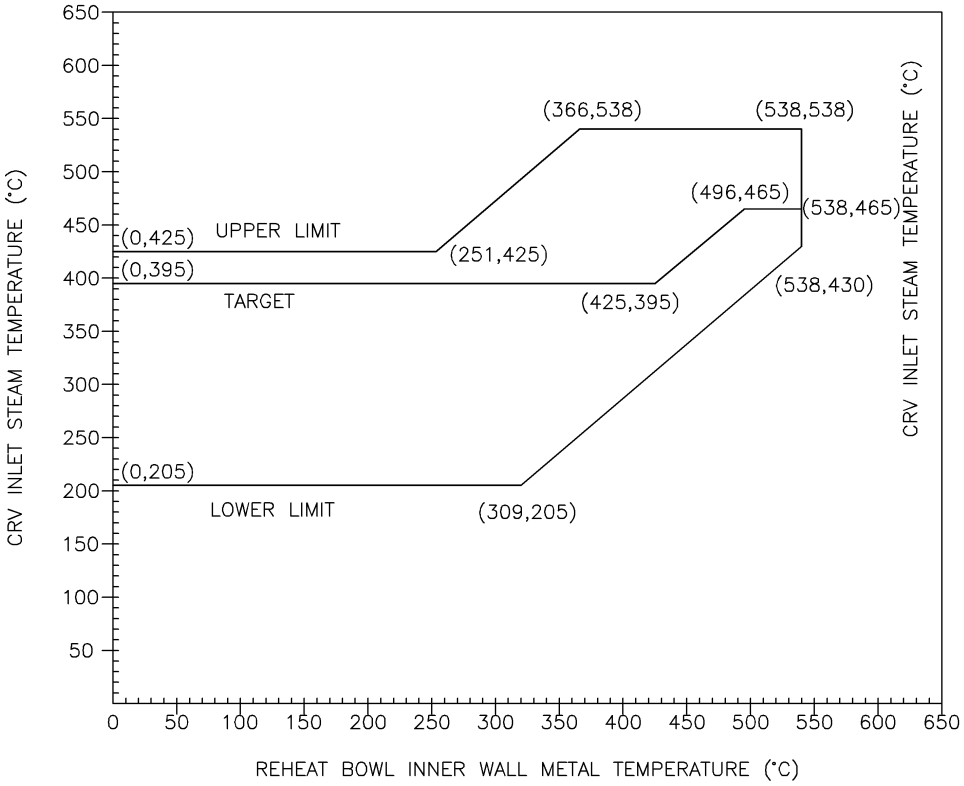


FIG. 3 ALLOWABLE TEMPERATURE DIFFERENCE ON REHEAT STEAM AND TURBINE METAL



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SERVICE	VARIABLE SET POINT						
Responsible dep.	Scale		DWG. No.	Rev.	Sheet	Date of issue	
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1		310SC90-776	0	21	2015-05-29	
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DWG. No. 310SC90-776

FIG. 4 ALLOWABLE TEMPERATURE DIFFERENCE ON MAIN STOP VALVE CASING

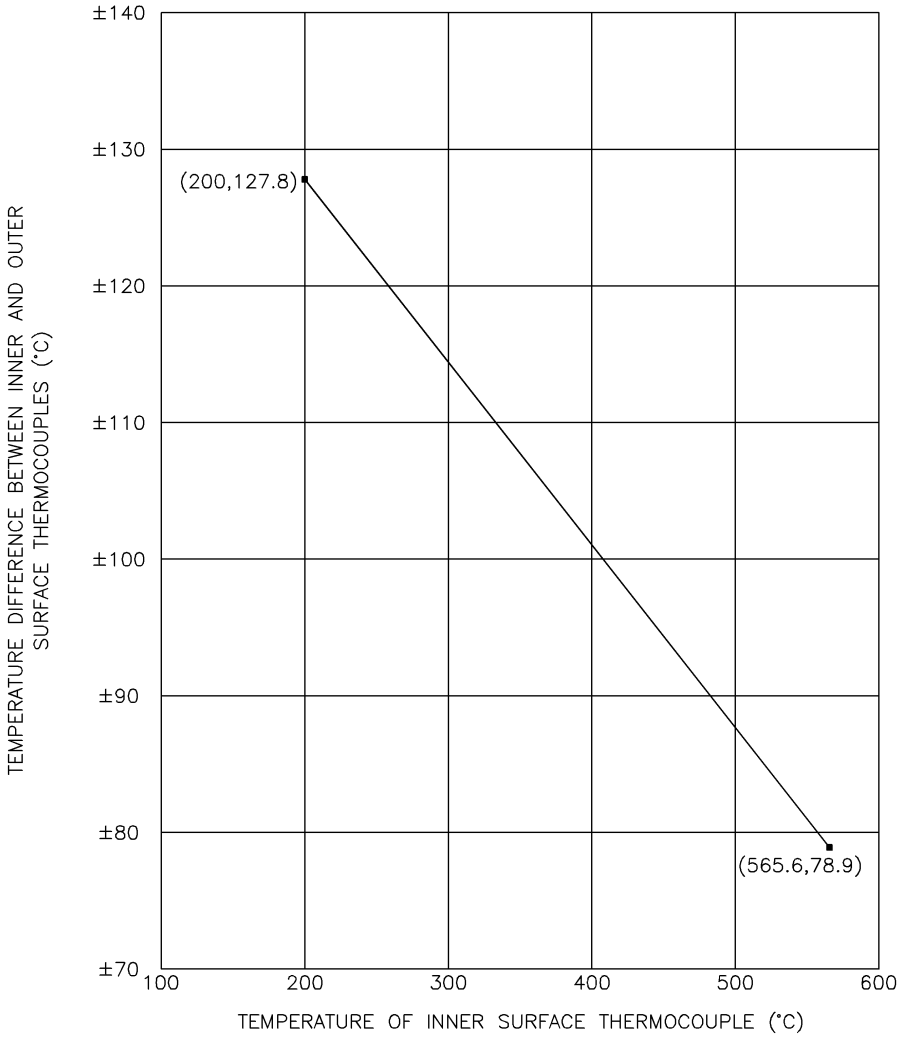
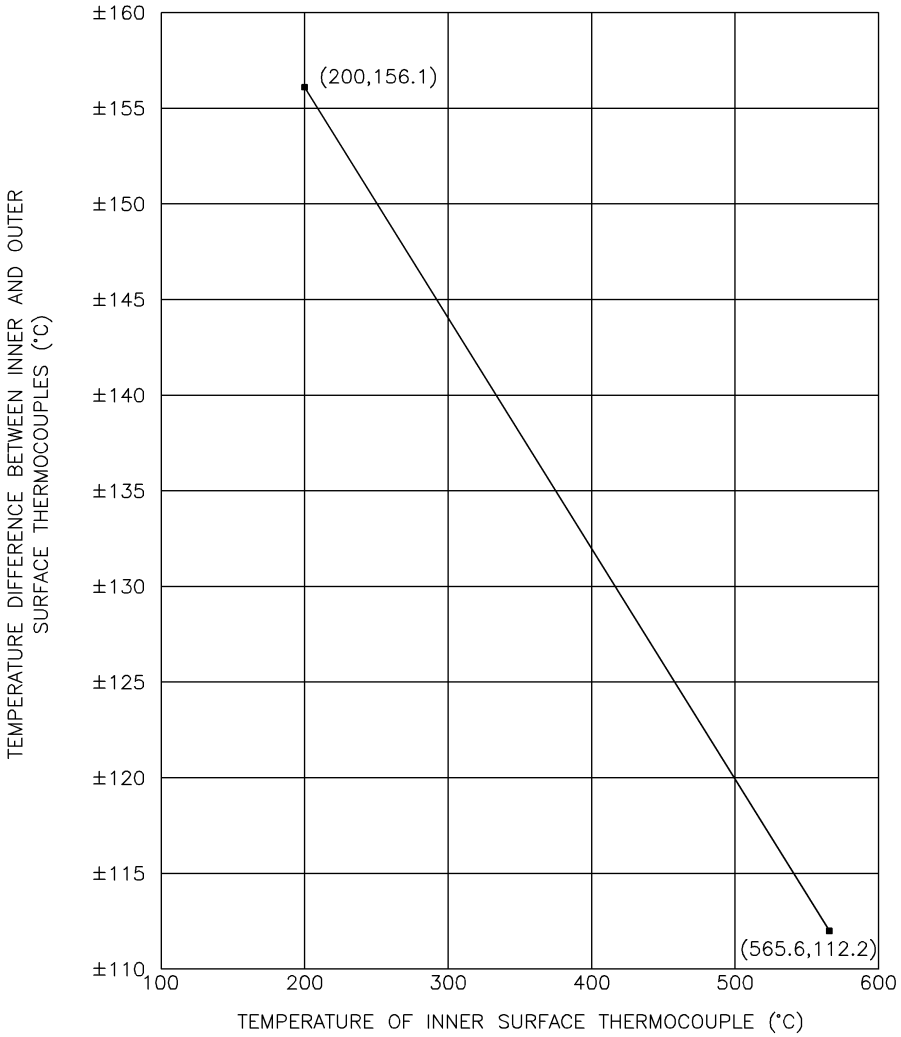


FIG. 5 ALLOWABLE TEMPERATURE DIFFERENCE ON TURBINE CONTROL VALVE CASING



TYPE	–					PAGE	22
SERVICE	VARIABLE SET POINT						
Responsible dep.	Scale		DWG. No.	Rev.	Sheet	Date of issue	
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FIG. 6 ALLOWABLE TEMPERATURE DIFFERENCE ON HP FIRST-STAGE SHELL

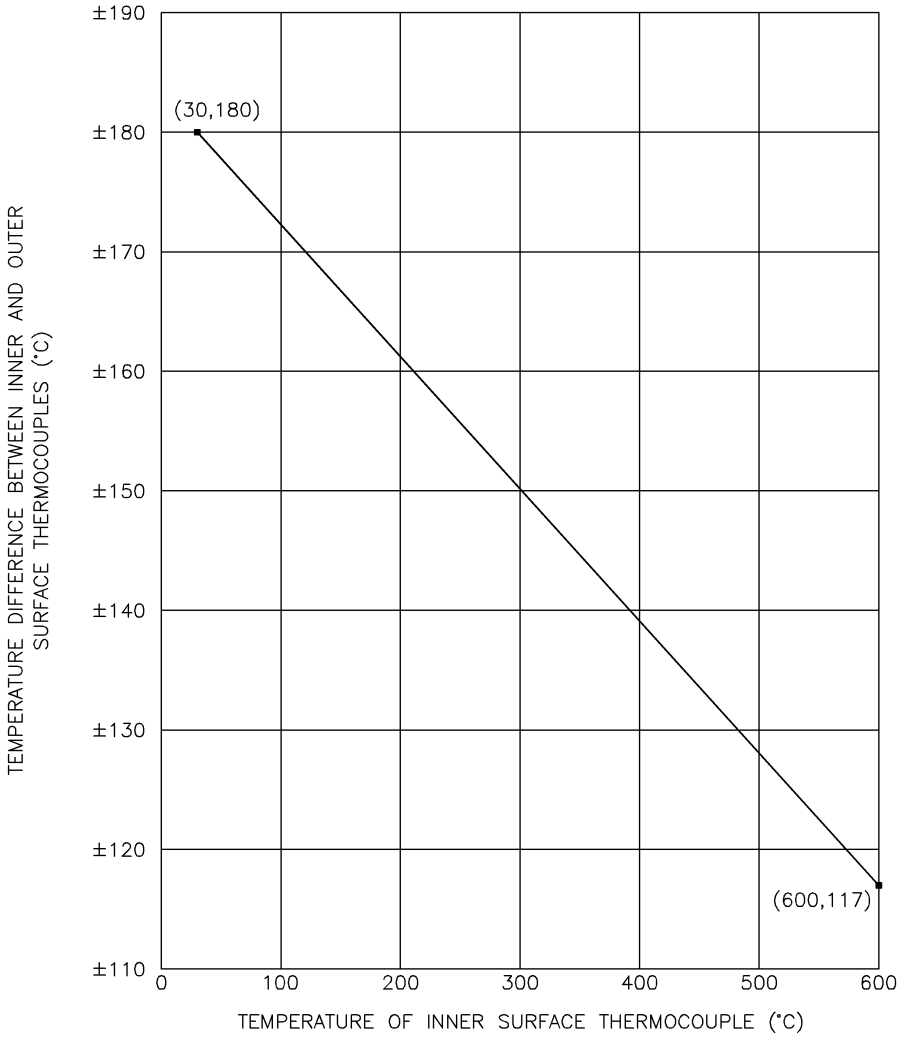
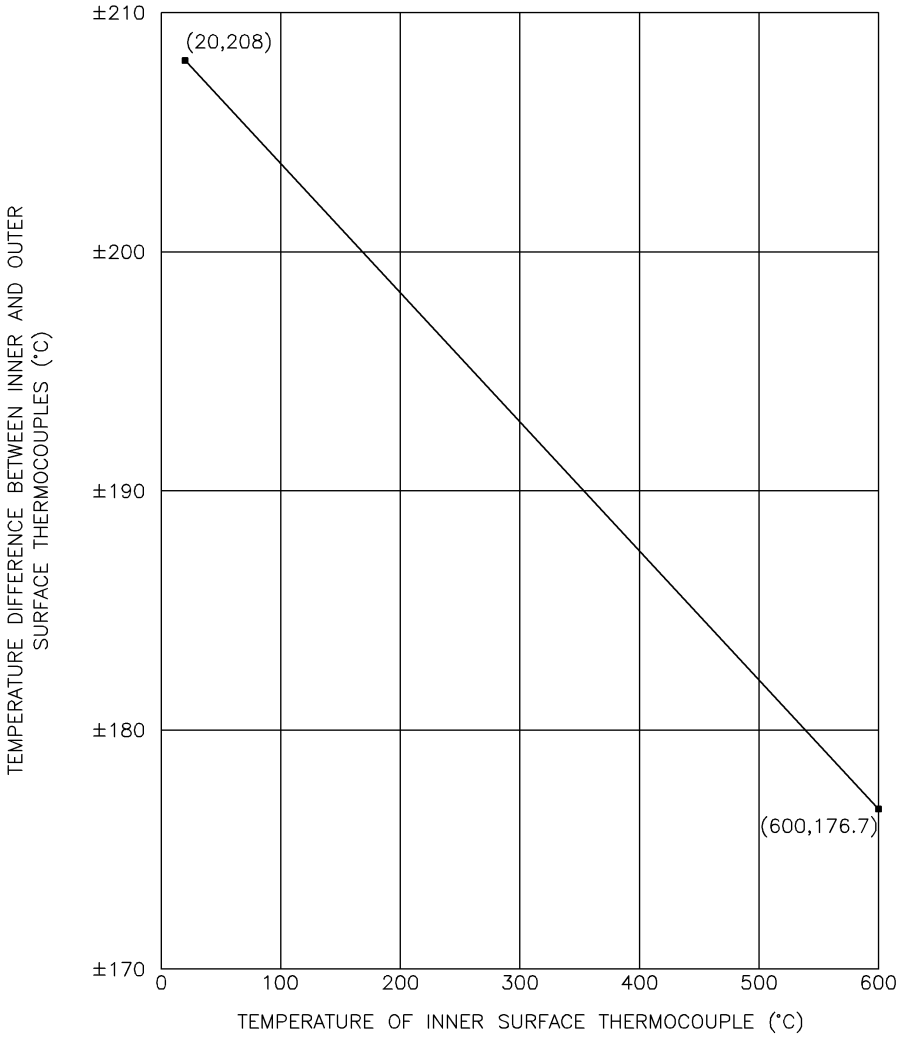


FIG. 7 ALLOWABLE TEMPERATURE DIFFERENCE ON REHEAT BOWL SHELL



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SERVICE	VARIABLE SET POINT						
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